

Name: Omaisma Rauf

Roll No: 198

Section: B1



Subject: Physiology

Grace collage of pharmacy
and health science khampur

Sr.No List of Experiment

- 1- Determination of coagulation time (CT).
- 2- Determination of bleeding time (BT) by Duke's Method.
- 3- Determination of blood group.
- 4- Measurement of blood pressure.
- 5- Determination of erythrocyte sedimentation Rate (ESR).
- 6- Testing of visual acuity (VA).
- 7- Recording and interpretation of ECG.
- 8- Determination of Red Blood Cell count.
- 9- Determination of total leukocyte count (TLC).
- 10- Determination of Hb in blood.



Experiment No 1:-

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Determination of coagulation time (CT)

Apparatus :-

- Needle
- Stop watch
- Capillary tube
- Cotton
- Spirit



Experiment No: 1

Determination of coagulation time (CT).

Introduction:-

The coagulation time is the time required for a sample of blood, removed from the blood vessel, to clot.

Normal value:-

2-6 min (by capillary tube method).

Apparatus:-

- Needle
- Stopwatch
- Capillary tube
- Cotton
- Spirit

Capillary tube method:-

Procedure:-

- i- Clean the finger with spirit swab and allow to it air dry.

Clotting Time:-



Add the blood and
start the timer



Record the time
when clot form

 Keep ~~on~~ breaking the tube every
30sec to see the clot formation.



- Puncture it with needle and discard the first drop of
- ii- Hold the capillary tube with horizontally. Allow the tube to fill by capillary action.
- iii- A small piece bit of the glass tube is broken off every 30 second untill a fine thread of clotted blood appears while the capillary tube is broken.
- iv- Repeat the step 3 times after 30 second untill such a thread is obtained.
- v- Record the coagulation time.

Precaution:-

- Avoid air bubbles in the capillary tube.
- Keep the other end of capillary tube open while filling it with drop of blood.

- The method should be repeated for 3 capillary tube to record accurate timing.



Experiment No 2:

★ Determination of bleeding time (BT) by
DUKE's method.

Apparatus:-

- Lancet
- Stopwatch
- Filter paper
- Cotton
- Spirit

Experiment No: 2

Determination of bleeding time (BT) by DUKE's method.

Introduction:-

The bleeding time is taken as the time required for a small shape incision to stop bleeding. The test helps in the diagnosis of various hemorrhagic disorder of vascular or platelets origin.

Apparatus:-

- Lancet
- Stopwatch
- Filter paper
- Cotton
- Spirit

Normal Value:-

1 - 6 min (by DUKE's method).

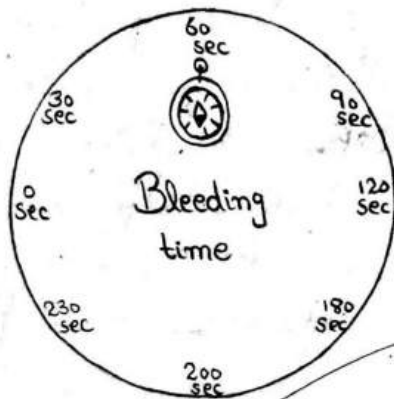


Fig : Bleeding time



Procedure:-

- i- Clean the finger tip with spirit swab and allow it to air dry.
- ii- Puncture the finger tip with lancet about **3mm** deep.
- iii- Set stopwatch and start it note that moment bleeding start from the puncture.
- iv- Apply the edge of filter paper on blood drop and take this time as zero reference.
- v- At **15 sec** interval wipe the blood drop, away completely with filter paper continue this process untill no more blood stain is picked up on the filter paper.
- vi- Record the difference of time between the starting and stoping of bleeding.

Precautions :-

- Puncture should be done under aseptic measure.
- Do not touch finger tip while wiping the blood away.

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Experiment No : 3

★ Determination of blood group.

Apparatus:-

- Anti serum A, B, D
- Glass slide
- Cotton swab
- Spisit
- Lancet

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Experiment No: 3

Determination of blood group.

Introduction:-

Blood groups are inherited characters. They are transmitted from parents to their offspring through genes/chromosomes.

Types:-

OAB or ABO blood groups.

ABO blood groups are further classified into :

O

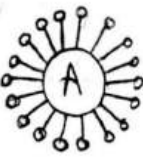
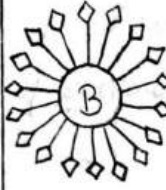
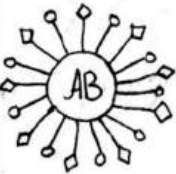
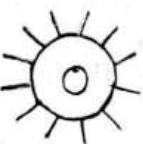
A

B

AB

Rh blood group system:-

- Rh +ve
- Rh -ve

	A	B	AB	O
Red blood cell type				
Anti-bodies in Plasma	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens in red blood cells	 A-antigen	 B-antigen	 A and B antigens	None

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Apparatus:-

- Anti serum A, B, D
- Glass slide
- Cotton swab
- Spirit
- Lancet

Procedure:-

- One drop of anti serum A was placed on one end of glass slide, one drop of anti serum B and one drop of anti serum D was placed on the other end of the slide.
- One drop of blood was dropped on each drop of anti serum which were labelled A, B and D.
- One drop of RBC suspension was mixed with antiserum.
- The absence of agglutination was conformed by the clear mixture with dispersed RBC's.

- The presence of agglutination is conformed by the presence of thickness of RBC's.
- For the determination of Rh factor, Anti serum D was mixed with RBC suspension. The presence of Rh factor Anti serum D was conformed by the presence of thick mass of RBC's.

Interpretation:-

- 1- If agglutination occur in anti-serum A the RBC's contain A antigen so the blood group is A.
- 2- If the agglutination occur anti-serum B the blood group is B.
- 3- If the agglutination occurs with both anti-serum A and B the blood group is **AB**.
- 4- If agglutination does not occur with both anti-serum A and B the blood group is **O**.

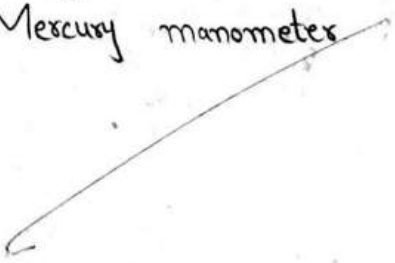
5- For the determination of Rh factor, Anti-serum D is mixed with RBC suspension. The presence of Rh factor Anti-serum D was confirmed by the presence of thick mass of RBC's.



Experiment No. 4

★ Measurement of blood pressure

Apparatus:-

- Stethoscope
 - Sphygmomanometer
 - Mercury manometer
- 

Experiment No : 4

Measurement of blood pressure:

Introduction:-

The blood pressure is the pressure exerted by the flowing blood against the walls of arteries.

Arterial blood pressure has two important component :-

- 1- Systolic blood pressure
- 2- Diastolic blood pressure

Normal value:-

It's normal value is $120/80$ mmHg.

Systolic blood pressure :-

The highest pressure exerted by the blood on the walls of arteries produced by the ventricular contraction.

It's normal value is 120 mmHg.

Blood pressure category	Systolic mmHg (upper numbers)		Diastolic mmHg (lower numbers)
NORMAL	Less than 120	and	Less than 80
ELEVATED	120 – 129	and	Less than 80
High B.P (hypertension) stage 1	130 – 139	or	80 – 89
High B.P (hypertension) stage 2	140 or higher	or	90 or higher
Hypertensive (consult your doctor)	Higher than 180	and/or	Higher than 120



Diastolic blood pressure:-

The lowest blood pressure exerted by the blood on the wall of arteries during ventricular relaxation.

Its normal value is 80 mmHg.

Apparatus:-

- Stethoscope
- Sphygmomanometer
- Mercury manometer

Procedure:-

- Cuff is tightly placed around the arm and chest piece of stethoscope is placed below the cuff over the antecubital fossa.
- Pressure is raised in cuff so that the brachial artery occluded due to compression.

- After that, arm cuff is deflated, while doing so series of sounds are heard through stethoscope.
- Appearance of clear tapping sound during first phase indicated systolic blood pressure and disappearance of sound is in 5th phase show diastolic blood pressure.

we note the value of systolic and diastolic blood pressure.

Precautions:-

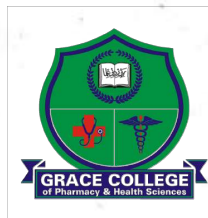
- i- Do not eat or drink anything 30mins before you take your B.P.
- ii- Sit in a comfortable chair with your back supported for at least 5 minutes before your reading.

Experiment No: 5

★ Determination of erythrocyte sedimentation rate (ESR).

Apparatus:-

- Spirit swab
- Syringe
- Anticoagulant
- Gloves
- Westergren tube
- Westergren stand



Experiment No: 5

Determination of erythrocyte sedimentation rate (ESR).

Introduction:-

In the presence of an anticoagulant the process of settling down of RBCs from the specimen of blood is known as sedimentation.

The rate at which this settling of RBCs occurs is known as the erythrocyte sedimentation rate.

Determination of ESR:-

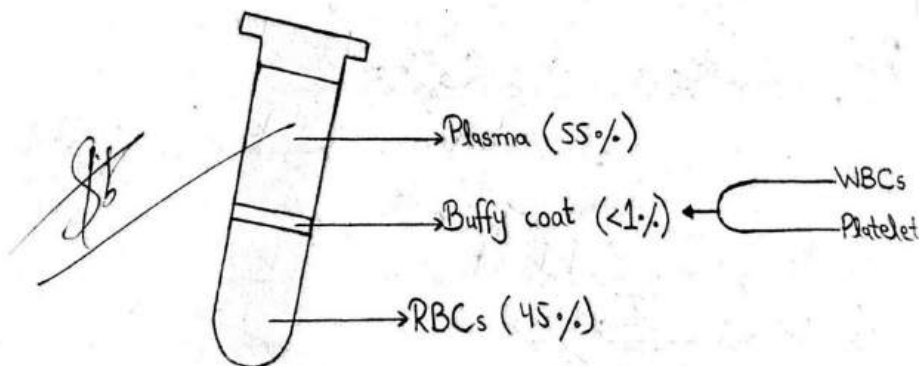
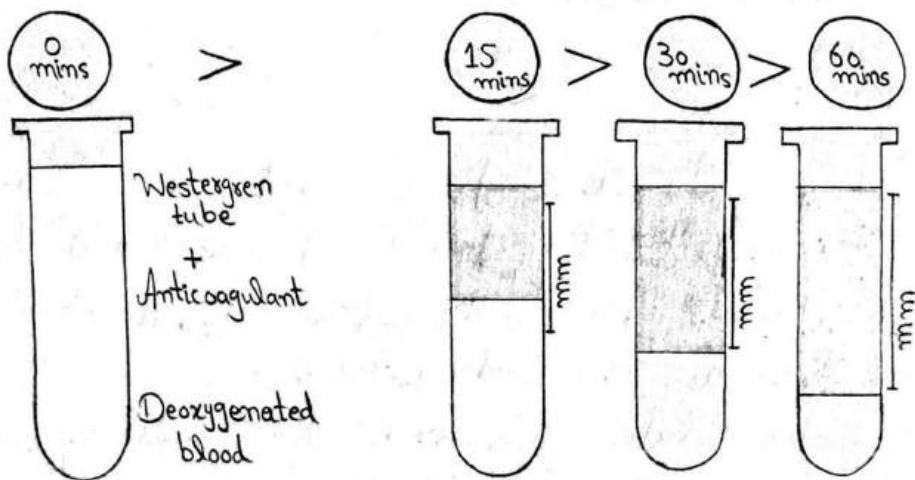
There are two methods to determine ESR:-

i- Westergren Method

Normal value:-

Males = 5-10 mm / 1st hour

Females = 10-15 mm / 1st hour



ii- Wintrob's method

Normal values:-

Males = 0-10 mm/
Females = 0-15 mm/
9th hour

Apparatus:-

- Spirit swab
- Syringe
- Anticoagulant
- Gloves
- Westergren tube
- Westergren stand

Procedure :-

- 1- Draw the blood from the vein of subject with the help of syringe.
- 2- 1.6 ml of blood is mixed with 0.4 ml of anticoagulant and filled in westergren tube.
- 3- The tube is fitted to the stand vertically and left

4 The reading is taken at the end of one hour.

Significance of determination of ESR:-

ESR helps in diagnosis as well as prognosis. ESR is increased in inflammation, anemia, fever, some kidney diseases and in cancer.



Experiment No 6

★ Testing of visual acuity

Apparatus:-

- Snellen's chart
- Subject



Experiment No: 6

Testing of visual acuity.

Introduction:-

Clarity or the sharpness of vision. Visual acuity is a measure of the ability of eye to distinguish shapes and the details of the objects at a given distance. Clinically visual acuity is often determined by use of the familiar Snellen's letter chart.

Apparatus:-

- Snellen's chart
- Subject

Procedure:-

For distant vision:-

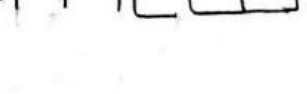
Subject being tested is 20 feet away from Snellen's chart

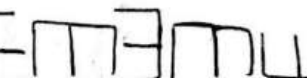
$\frac{20}{200}$ E $\frac{200\text{FT}}{61\text{M}}$ 1 | D-30,0  V-0,1 1

$\frac{20}{100}$ F P $\frac{100\text{FT}}{30,5\text{M}}$ 2 | D-20,0  V-0,2 2

$\frac{20}{70}$ TOZ $\frac{90\text{FT}}{27,3\text{M}}$ 3 |  V-0,3 3

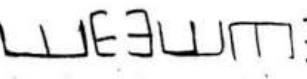
$\frac{20}{50}$ LEPD $\frac{50\text{FT}}{15,2\text{M}}$ 4 | D-16,75  V-0,4 4

$\frac{20}{40}$ PECFD $\frac{40\text{FT}}{12,2\text{M}}$ 5 |  V-0,5 5

$\frac{20}{30}$ EDFCZP $\frac{30\text{FT}}{9,1\text{M}}$ 6 | D-12,5  V-0,6 6

$\frac{20}{25}$ FELOPZD $\frac{25\text{FT}}{6,1\text{M}}$ 7 | D-10,0  V-0,7 7

$\frac{20}{20}$ DEFPOTEC $\frac{20\text{FT}}{6,1\text{M}}$ 8 | D-8,33  V-0,8 8

$\frac{20}{15}$ LEFODPCT $\frac{15\text{FT}}{4,5\text{M}}$ 9 | D-7,14  V-0,9 9

$\frac{20}{10}$ FDELTKD $\frac{10\text{FT}}{3,0\text{M}}$ 10 | D-6,25  V-0,10 10

$\frac{20}{5}$ PELOLCDF $\frac{10\text{FT}}{3,0\text{M}}$ 11 |  V-0,11 11

and read smallest line.

- Snellen's charts are designed so that a normal individual can read the letter in the smaller line at 20 feet or 6m.
- A person visual acuity is stated as $V = \frac{d}{D}$ in which "d" is the distance at which the patient can read the letter and "D" is the distance at which the normal eye can read the letter.
- If vision is recorded $10/20$ it is subnormal because one must approach to within 10 feet to read letter that are readable at 20 feet by the normal eye.

For near vision:-

- Ability to see clearly an image that is relatively close - (about 10-13 inches)

- Near vision is usually measure with a hand held card similar to the eye chart on wall.
- Font size is characterized from N₄-N₂₄.
- N₆ in the near vision is equivalent to 20/20 in Snellen's chart.
- The smallest point used his N₅.

Precation:-

1. Examiner must have 6/6 vision and no color blindness.
2. Each eye should be tested separately.



Experiment No 7

Reading and interpretation of ECG.

Apparatus:-

- ECG machine
- Electrode gel
- Subject

Experiment No : 7

Recording or Interpretation of ECG.

Introduction:-

Electrocardiography is the technique of recording the electrical activity of the heart. The recording itself is called an electrocardiogram.

Composition:-

A normal ECG is composed of a :

P wave

QRS complex

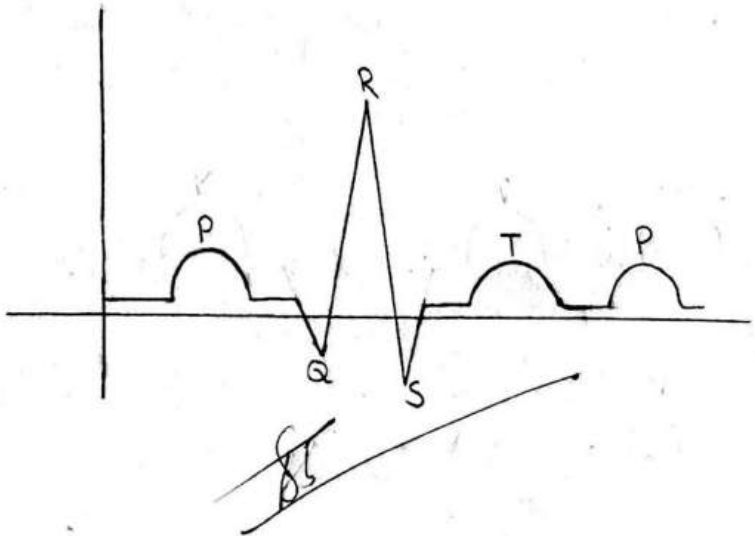
T wave

Apparatus:-

- ECG machine
- Electrode Gel
- Subject

Electrocardiography (ECG)

Recording and Interpretation



Procedure:-

- Connected the limb and chest electrode properly using contact gel. Make sure that the subject is comfortable and wearing no metallic objects.
- Check the amplitude on QRS complex in lead I, II and III. According to the Einthoven's Law.
- Examine the initial strip of the record and check that there should be no extra line.
- Record in the sequence first for standard limb lead followed by augmented limb lead.
- After having record the ECG calculate the heart rate first and then observe the various components of ECG carefully.

Precautions:-

1. Relax the patient before taking

a record.

2. Use proper voltage and handled the electrical record gently.
3. There should be no metallic objects on the subject.

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Experiment No: 8

★ Determination of Red blood cell count.

Apparatus:-

- RBC's pipette
- blood lancet
- Cover slip
- Spirit swab
- NaCl = 1gm
- Na_2SO_4 = 5gm
- Water = 200ml
- Neubaur's chamber

Experiment No : 8

Determination of Red blood cell count.

Introduction:-

The RBC's count is almost always part of a complete blood count, CBC is a group of test that measure many different part and features of blood.

Normal value:-

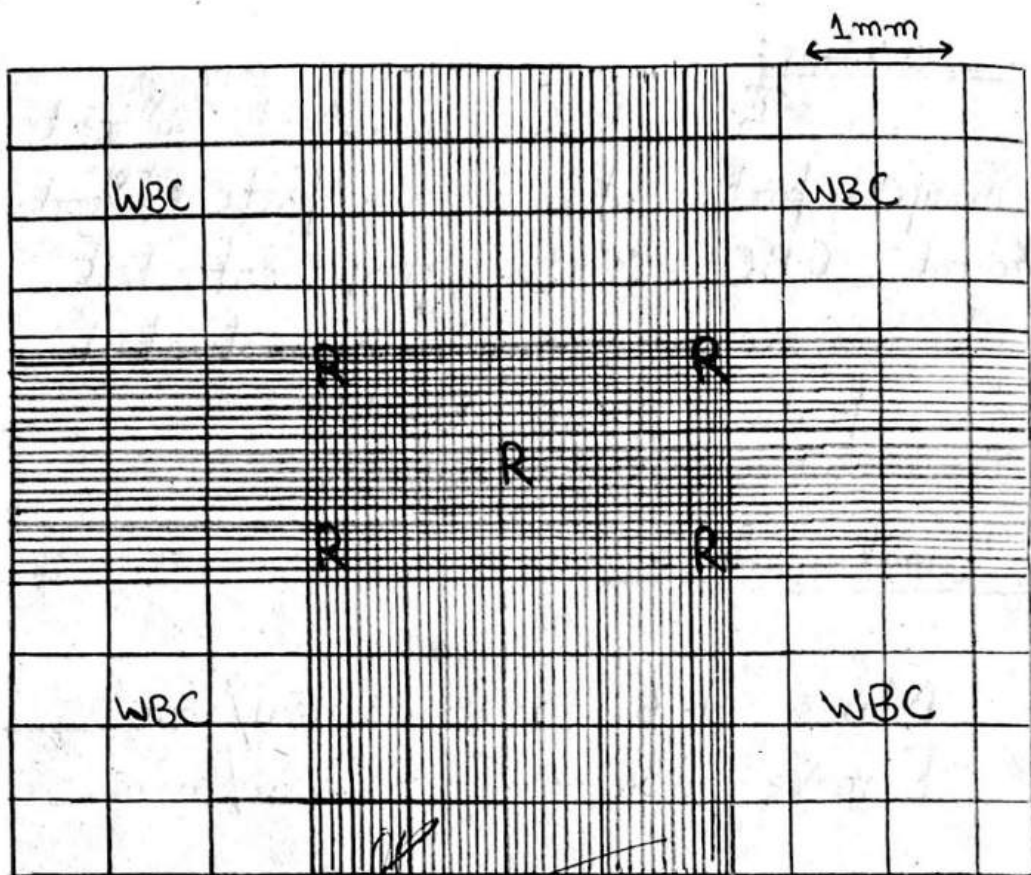
Males 4.6 — 6.0 million/mm³

Female 4.0 — 5.2 million/mm³

Apparatus:-

- RBC's pipette
- blood lancet
- Cover slip
- Spirit swab
- NaCl = 1gm
- Na₂SO₄ = 5gm





$$\boxed{\text{RBC/cmm}} = \text{no. of cell counted} \times \text{dilution factor} \times \text{Depth} \times \text{Area counted}$$

$$\boxed{\text{RBC/cmm}} = 500 \times 200 \times 10 \times 5 = 5,000,000$$

- Water = 200ml
- Neubaur's chamber

Procedure:-

- Draw and mix the blood in RBC pipette up to 0.5
 - Now draw the diluting fluid up to 120 mark.
 - Gently rotate the pipette between thumb and finger.
 - Alternatively prepared $1/200$ dilution of blood by mixing.
 - Place the cover slip on Neubaur chamber.
 - Discard first 4 to 5 drops from RBC's pipette before charging Neubaur chamber.
- Wait for two minutes to allow the RBC's cell to settle.

- Count the RBC's at the central large square and four other corner count at last 500 cells.

Precautions:-

- Select the correct position of Neubaur's chamber.
- There should be no overflowing of fluid in the H groove of chamber.



Experiment No : 9

★ Determination of TLC.

Apparatus:-

- WBC's Pipette
- Blood Lancet
- Covers slip
- Glacial acetic acid = 1ml
- Water = 100ml
- Neubauer's chamber

Experiment No: 9

Determination of TLC.

Introduction:-

A WBC count is a blood test to measure the number of white cell in the blood.

WBC's are also called leukocytes. They help to fight against infection.

Classification:-

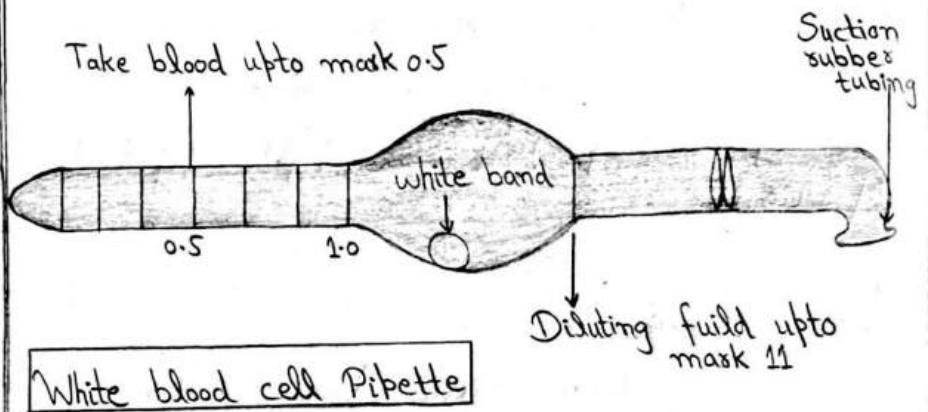
There are two major classes of WBC's:

i Granulocyte:-

- Basophills
- Eosinophills
- Neutrophills

ii Agranulocyte:-

- Monocytes
- Lymphocytes



Normal value :-

4000 - 11000/mm³

Apparatus:-

- WBC pipette
 - Blood lancet
 - Cover slip
 - Glacial acetic acid = 1ml
 - Water = 100ml
- Neubaur's chamber

Procedure :-

- First sterilize the hands and prick the finger of subject.
- Take WBC pipette and suck the blood from the pricked finger upto 0.5 mark.
- Add the diluting fluid glacial acetic acid upto 11 mark into the WBC pipette.
- Roll WBC pipette with your palms.

- v. Now Discard the 1 mark, the stem part of the pipette, as this contains the diluting fluid.
- vi. Place the cover slip on Neubaur's chamber, add 1 drop on each side of coverslip so it spreads well in the Neubaur's chamber.
- vii. Examine the Neubaur's chamber on the microscope under 10x lens. Add count all the WBC's in the four corner.

Precautions:-

- Select the correct portion of Neubaur's chamber.

There should be no overflowing of fluid in the H groove of chamber.

Experiment No : 10

★

Determination of (Hb) in the blood.

Apparatus:-

- Hemoglobinometer
- Lancet
- Diluting tube
- Hb pipette
- HCl solution



Experiment No: 10

Determination of (HB) in the blood.

Introduction:-

Hemoglobin is a red, Oxygen carrying pigment present in RBC's. It is consist of 4% heme and globin 96%.

The amount of Hb in the blood is measured by destroyed the red cell membrane, so that the pigment is released into the plasma and can be estimated.

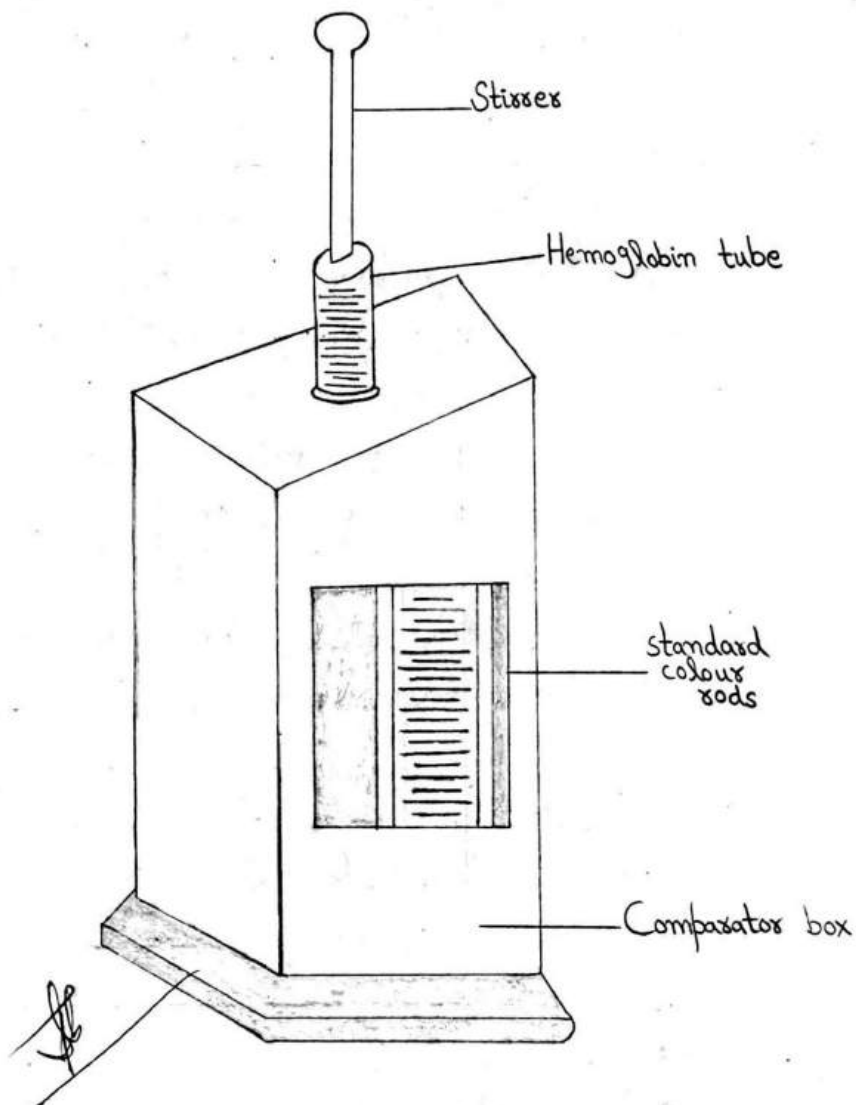
Normal value:-

Males = 14 - 16 gm/100ml

Females = 12 - 14 gm/100ml

Apparatus:-

- Haemoglobinometer
- Lancet
- Diluting tube



- Hb pipette
- HCl solution

Procedure:-

- Sahli's hemoglobinometer is a manual device that contains Hb tube, pipette and stirrer as well as a comparator.
- Hydrochloric acid converts hemoglobin to acid hematin which is then diluted until the color of the solution matches that of the comparator block.
- Note the reading on Hb tube at which color of blood matches with the comparator block.
- This is one of the most common methods for hemoglobin determination in developing countries and is relatively simple.

Precautions:-

- Fill the pipette with the blood.
- An appropriate Hb pipette must be selected.
- The hemoglobinometer must be cleaned with distilled water.

